

Evaluating a Neural-Network-Based Pressure-Estimation Model as a CFD Surrogate

9/14/2026

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Common Abreviations:

ARC	American Rocketry Challenge
CFD	Computational Fluid Dynamics
CG	Center of Gravity
CP	Center of Pressure
FAA	Federal Aviation Administration
NAR	National Association of Rocketry
NASA	National Aeronautics and Space Administration
PLA	Polylactic Acid
SDS	Safety Data Sheet
SLI	Student Launch Initiative
STEM	Science, Technology, Engineering, and Math
TRA	Tripoli Rocketry Association
PDR	Preliminary Design Review
CDR	Critical Design Review
FRR	Flight Readiness Review
PLAR	Post-Launch Assessment Review
AGL	Above Ground Level
ADC	Analog-to-Digital Converter
AHJ	Authority Having Jurisdiction
APCP	Ammonium Perchlorate Composite Propellant
ATC	Air Traffic Control
CAD	Computer Aided Design
CFR	Code of Federal Regulations
DOT	Department of Transportation
GPS	Global Positioning System
ID	Inner Diameter
IMU	Internal Measurement Unit
KE	Kinetic Energy
NAR	National Association of Rocketry
OD	Outer Diameter
PCB	Printed Circuit Board
PPE	Personal Protective Equipment
RSO	Range Safety Officer
WBS	Work Breakdown Structure

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1 General Information

1.1 About the Team

Washington Youth Rocketry, informally known as Team Goat, is a student rocketry team based in Bellevue, Washington. The team brings together six students from Interlake High School who share an interest in aerospace engineering, incorporating hands-on design, testing, teamwork, and competition. The six students making up the team first developed their interest and experience in rocketry through Odle Middle School's rocketry club, where they competed in the American Rocketry Challenge as members of two separate teams.

As the students moved on to Interlake High School, they continued working together and combined their previous rocketry experience to form a unified team. After qualifying for the National Finals, the team finished second nationally, placing them among the top student rocketry teams in the country. The team got the opportunity to represent the United States at the International Rocketry Challenge at the Farnborough International Airshow, where they won the competition.

This year, our team plans to compete in NASA Student Launch (SLI) to further develop our aerospace engineering skills while taking on more demanding engineering challenges.

1.2 Adult Educators Information

Name: Linhbao Tran
Email: linhbaotran@gmail.com
Phone Number: +1 (206) 376-8853

Name: Mehul Mediwala
Email: mmediwala@hotmail.com
Phone Number: +1 (425) 503-9543

1.3 Team Mentor Information

Name: Brendan Williams
Title: CTE teacher at Odle Middle School
Email: s-williamsb@bsd405.org
Phone Number: +1 (206) 349-7649
NAR Number: 97129
Certification: Level 2

1.4 Team Roles

Team Lead:
Name: Antoine Vigneron
Email: anhkhoa.vigneron@gmail.com

Safety Officer:
Name: Daniel Tadesse

Our team will be comprised of six members this year. Each member's responsibilities are outlined below.

Anay Mediwala: Payload Hardware, CFD	**Antoine Vigneron: Payload Hardware, Payload Software**
Anay will specialize in **computer-aided design** of flight bays and utilization of CFD software.	Antoine will specialize in flight electronics, payload construction, and state estimation.
Daniel Tadesse: Safety, Construction	**Mikhail Antipin: Payload Software, Payload Hardware**
Daniel will specialize in recovery systems**, machining,** and safety.	**Mikhail will specialize in the neural network and data quality.**
Bao-Ky Tran: CFD, Data Analysis	**Nikhil Sirivara: Safety, Construction**
Bao-Ky will specialize in CFD software and **quality control of** the **simulations**.	Nikhil will specialize in the construction of the main rocket body, safety, and motor dynamics.

Each team member will take on shared roles and responsibilities in parts of the team. Each person is expected to contribute to documentation of their respective responsibilities. All members must participate in launches, where everyone will have jobs in preparing the rocket, recovery and payload.

1.5 NAR/TRA Section

We plan to work with either Washington Aerospace Club, Tri-Cities Rocketeers, or Gorge Rocket Club:

- *Washington Aerospace Club* is both affiliated with the NAR and TRA. It is NAR Chapter #578 and TRA Prefecture #41.
- *Tri-Cities Rocketeers* is affiliated with both NAR and TRA. It is NAR Chapter #736 and TRA Prefecture #156.
- *Gorge Rocket Club* is affiliated with NAR. It is NAR Section #790.

Each club provides high-power rocket launch sites in Eastern Washington, a several-hour drive from Bellevue, where we are located. In addition, Washington Aerospace Club governs 60 Acres Park, a large field half an hour away, which we will use for our subscale launches.

1.6 Design Review Presentation Timeslots

All times are listed in Central Time

- **Wednesday 3-4 pm**
- **Wednesday 2-3 pm**
- **Friday 4-5 pm**

1.7 Letter of Administrative Support

Washington Youth Rocketry

Dear NASA Student Launch Management Team,

I, Andrew Shi, Executive Director of the Washington Youth Rocketry organization, am writing to make it clear that I fully support our student rocketry team in their participation in the NASA Student Launch Initiative for 2026/2027.
We are fully conscious of the engineering, academic, and safety milestones required by this program, and are confident in our organization's participation.
We are also comfortable with the students spending necessary time to meet for the project and the mandatory NASA milestone presentations, such as the Preliminary Design Review (PDR), the Critical Design Review (CDR), and the Flight Readiness Review (FRR).
We highly value the STEM opportunities offered by NASA and are looking forward to supporting our students as they take part in this national challenge on behalf of Washington Youth Rocketry.

Sincerely,
Andrew Shi
Executive Director | Washington Youth Rocketry
office@wayouthrocketry.org

2 Facilities/Equipment

2.1 Meeting Locations

We will meet multiple times each week. These meetings will take place in-person at our workshop at Antoine Vigneron's house, any team member's house for non-rocket tasks, or online.

Some meetings will use smaller groups so the team can complete independent tasks in parallel.

Table 2.1.1: Meeting Locations

Location	Description
Workshop	The workshop at Antoine Vigneron's house will serve as a place to work on the rocket. We will have a large workspace to perform major glue-ups and allow adhesives to cure overnight. It will be open most of the week.

Team Member House	For work on presentations, payload, or other team tasks that we want to do in person, various team members' houses will serve as a meeting location.
Online	Most meetings to work on documentation or to discuss and collaborate on ideas and online tasks can be completed through Teams or FaceTime. Online meetings are typically more suitable for frequent meetings because they are convenient.

2.2 Meeting Hours

The workshop will be available every day to meet. We will also meet each Sunday from 2-5 pm.

2.3 Necessary Personnel

For our workshop, we plan to implement a two-person work structure. This means that at least two people must be present whenever work is being performed on the rocket. This structure is intended to improve safety, efficiency, and the overall quality of our work for several reasons:

- Leads to better decision-making making as it allows the team to discuss decisions and consider multiple perspectives.
- Improves efficiency because members can divide tasks and assist one another
- Allows for higher-quality work as working in pairs or groups helps to catch mistakes, verify procedures and provide assistance.
- Results in increased safety as it allows the team to recognize different potential hazards when working and respond appropriately.

If not working in the shop, for example on the payload, one person is acceptable due to there being lower risks. However, major decisions should be taken with respective sub-team members.

2.4 Equipment

Inside our workspace, we will have a large table to work at, providing clear and open space to limit clutter and possible safety hazards. Equipment readily available includes smaller handheld tools such as hammers, hacksaws, wrenches, screwdrivers, files, sandpaper, and clamps. Higher power tools requiring specific safety precautions outlined in their use manuals like Dremel rotary tools and jigsaws are to be secured inside the workshop to prevent unsanctioned use. Finally, for ventilation a rotary vane vacuum pump will be utilized when deemed necessary for safety during construction.

2.5 Supplies

An initial parts list needed to build and design the rocket is detailed in Table 5.2.1. Currently, the team owns certain materials from accumulated rocket experience, such as two PerfectFlite Pnut altimeters, one Arduino Nano, and two Arduino Uno boards. Additionally, to meet the required safety standards, the team will purchase general PPE, including safety glasses, cut resistant nitrile gloves, and latex gloves throughout the year.

Additional programs will mainly be utilized through free, accessible to students, or opensource software. This software includes applications through Microsoft Office Suite, OpenRocket, OpenFOAM, and Onshape. Though the team additionally has access to RocketSim through vendor connections.

3 Safety

3.1 Safety Organization and Responsibilities

Roles:

- Safety Officer:
  - Responsible for shop safety, inspections, hazard log etc.
  - Oversees safety sections in documentation
  - Enforces protocols and briefs team on safety protocol
- NAR/TRA Mentor: Motor custody, design and build oversight, owner of rocket, present during ANY handling of motors.
- RSO: Final authority at the range, approves or denies launches.
- All Members: Follow procedures with safety as a priority, stop unsafe work, report hazards.

3.2 Hazard Analysis

3.2.1 Hazard Identification

Hazards will be anticipated and identified in advance by following a standardized procedure:

Hazards will be analyzed during:

- Design
- Fabrication
- Component Testing
- Integration
- Motor Preparation
- Transportation
- Launch Preparation
- Launch
- Recovery

Hazards will be anticipated by using sources such as:

- Manufacturer documentation
- SDS Sheets
- NAR/TRA Guidance
- Mentor Inspection

Any team member must immediately report a hazard to the Safety Officer. The Safety Officer will classify the hazard and approve mitigation before hazardous work begins. Hazard analysis will be performed by the safety officer in anticipation of each major activity, along with passive hazard observations by the team members.

All hazards will be logged in a common Excel spreadsheet managed by the Safety Officer to systematically verify that mitigations are implemented. The NAR mentor will review hazards related to the use and transportation of dangerous materials such as the motors.

3.2.2 Hazard Classification

This is an FMEA risk matrix. We chose this approach as it provides a comprehensive way to classify risk and is can provide insight into a large variety of possible hazards, including shop safety and launches.

The matrix works by assigning hazards a risk factor. Each risk factor is categorized through two metrics, severity and probability:

Severity × Probability = Risk Factor

Table 3.2.1: FMEA Risk Matrix

		Level of Severity			
		1	2	3	4
Probability of Occurrence	1	1	2	3	4
	2	2	4	6	8
	3	3	6	9	12
	4	4	8	12	16

Using this risk matrix, we will classify the risks for this year and outline the mitigation strategies to address them.

The probability of occurrence is split into four sections of 25% probability. Probability is ranked from 1-4, with 1 being the least probable, to 4, the most probable.

Table 3.2.2 FMEA Probability Breakdown

Probability of Occurrence	1	0-25% chance of occurrence
	2	25-50% chance of occurrence
	3	50-75% chance of occurrence
	4	75-100% chance of occurrence

Severity is split into four categories, measuring the impact on people, equipment, facilities, and rocket. Severity is ranked from 1-4, with 1 being least severe, to 4, the most severe.

Table 3.2.3 FMEA Severity Breakdown

Level of Severity			
1	2	3	4
Almost no consequences, small injury requiring no action, barely any damage	Less severe consequences, treatable injury, or repairable damage	Severe consequences, injury, or large damage	Major consequences, injuries, or unfixable damage

Risks 6 and above will not be tolerated, and any activity deemed to be of that risk or higher will require mitigation.

3.2.3 Hazard Mitigation and Tracking

When a hazard is identified, its risk will be classified using Tables 3.2.1–3.2.3. Depending on its severity, the Safety Officer will adequately come up with mitigation procedures and consult with the NAR mentor if necessary. With appropriate mitigation procedures, the risk will be re-evaluated.

This process will be documented in a team safety Excel spreadsheet to ensure that all risks are accounted for and properly addressed, including residual risks.

3.3 Preliminary Hazards Analysis

The following table presents a preliminary analysis of three hazards and mitigation strategies using our risk analysis procedure:

Table 3.3.1: Preliminary Hazard Analysis

Hazard	**Cause**	**Effect**	**Risk**	**Procedure**	**Mitigated Risk**
Personnel Hazards					

Cyanoacrylate bonding of skin or fingers.	Improper management and control of the glue (spills, slippage, seepage), lack of proper safety equipment (gloves).	Skin bonding or tearing, potentially thermal burns.	8	Nitrile gloves, safety glasses, and careful, unobstructed handling of the material.	1
Failure Modes					
Parachute gets tangled, and fails to deploy	Improperly packed parachute leading to tangles	Ballistic/uncontrolled deployment, risk of rocket breaking	6	Parachute will be folded according to procedure, and the packing strategy will be tested during ground deployment testing. In addition, an observer will validate that the parachute has been packed correctly.	3
Environmental Hazards					
Dry vegetation near launch site	Failure to observe dry vegetation combined with hot motor exhaust and debris and comply with safety regulations	Risk of uncontrolled fire, burns, smoke exposure, and property damage.	6	Systematic observation of launch grounds within a radius of 37.5 feet (NAR High Power Safety Code). Combustible materials in the area will be removed or avoided depending on feasibility.	3

3.4 Safety by Activity and Location

This section elaborates on how we will manage safety during design, fabrication, storage and launching.

3.4.1 Design

Structural margins such as weight loads or static margin of the design will be calculated using built-in tools in OpenRocket.

3.4.2 Fabrication

When using machines for construction, such as saws, rotary tools, 3D printers etc., members will be required to:

- Receive training before use, as dictated by OSHA standards
- Wear appropriate eye protection
- Wear closed toed shoes
- Have secured hair and clothing & remove jewelry
- Have tool guards in place
- Work in pairs as per the two-person work structure
- Immediately remove damaged equipment from use
- Adult oversight during machine operation
- Ensure full stop of moving parts before adjusting
- Use dust extraction system when appropriate
- Accessible fire extinguishers

When using hazardous chemicals, such as epoxy, CA glue, paint etc., members will be required to:

- Review SDS before use
- Use gloves or eye protection as specified
- Ensure proper ventilation
- No consumption of food or drink
- Dispose of waste according to SDS or local rules

We will also have a First-Aid kit available in the workshop in case of any injury. This First-Aid kit includes bandages, gauze pads, antiseptic ointment, alcohol swabs, and more to deal with any injuries.

Table 3.4.1: Workshop Tools Safety

Tool	Safety Procedure
Hand Tools (hammer, hacksaws, files, pliers, hand drill, etc.)	Ensure adequate eye, hand, and clothing protection is worn. Tools must be used on a hard and steady surface. At least two people working with the tools.
Soldering	Proper eye and hand protection must be worn. Ventilation system must be present. Secure soldering iron stand. Wet material for cooling and cleaning. Clamps for holding materials.
Rotary Tool	Ensure adequate eye, hand, and clothing protection is worn. Tools must be used on a hard and steady surface. At least two people working with the tools.

Power Sander	Ensure adequate eye, hand, and clothing protection is worn. Tools must be used on a hard and steady surface. At least two people working with the tools.

3.4.3 Storage

Chemical Storage:
Chemicals will be stored in their original labeled containers and in an appropriate and organized cabinet. Incompatible materials will be separated. SDS documentation will be readily accessible and stored nearby to review when the chemicals used.

3.4.4 Launches

Launch Sequence:

Before launch operations, the team will ensure compliance with FAA Part 101 Subpart C regulations.

Table 3.4.2: Launch Safety Table

Launch Phase	Safety & Compliance Action
Pre-Launch Operations	Check weather
	Check wind
	Check fire conditions
	Understand the launch site boundaries
	Inspect the rocket
	Obtain mentor approval
	Submit the rocket for inspection by RSO
	Safety briefing including all the above and below
At the Launch Pad	Secure the rocket appropriately on the approved launch rod
	Launch electronics remaining safe until armed
	Motor igniter will be installed only immediately before launch
	Deployment charges remain safely inert until immediately before launch
During Launch	All people will be cleared to the appropriate distance (200 feet for K class)

	Final RSO approval will be obtained
	A clearly audible countdown will be made
	Airspace will be ensured to be clear
	Flight will be aborted if suddenly not compliance with NAR weather regulations or airspace unclear.

In the event of a misfire, members will follow NAR safety guidelines by disconnecting the launch system's battery and waiting 60 seconds before investigating.

For recovery, the team will:

- Enter the range only after authorization from the RSO/LCO
- Not attempt to recover the rocket from power lines, dangerous trees, roadways, water, or another location where recovery would create an unreasonable risk
- Remaining energetic systems will be treated as armed until verified safe

60 Acres Park: Subscale launches will take place at 60 Acres Park, on the North field. This field has a ceiling cap of 2,000 feet. For this reason, our subscale launches can take place here, but not full scale. We will follow similar operating procedure at 60 Acres to previous years for ARC. This includes:

- Having a blast deflector on the launch rod, so as not to ruin the grass turf used for soccer.
- Check to see if the sky is clear of any aircraft before launching.
- Following all steps in the launch checklist.
- Launching near the middle of the field, moving to take in account wind. This will prevent the rocket drifting off-course and landing on the road, trees, river or forest nearby.

3.5 Safety Data Sheet Links

Table 3.5.1 Safety Data Sheet Links

JB SteelWeld	https://s3.us-west-2.amazonaws.com/catsy.744/8265s_J-B+Weld+Original+Steel+Reinforced+Epoxy+Twin+Tube_US+SDS+EN_OCT++2024.pdf
JB ClearWeld	https://s3.us-west-2.amazonaws.com/catsy.744/50112_50240H_ClearWeld+Syringe_US+GHS+SDS+EN_SEP+2025.pdf
BSI Insta-Cure CA Glue	https://bsi-inc.com/sds_pdf/sds_insta_cure.pdf
BSI Insta-Cure+ CA Glue	https://bsi-inc.com/sds_pdf/sds_insta_cure_plus.pdf
BSI Insta-Set CA Accelerator	https://bsi-inc.com/sds_pdf/sds_insta_set.pdf

3.6 Motor and Energetics Logistics and Safety

Motor logistics and safety will be subject to oversight at all times by our mentor.

- Motors will be purchased by the team mentor from established high-power rocketry vendors. Before purchase, the mentor will verify that the selected motor is commercially manufactured, NAR/TRA-certified, permitted by NASA Student Launch requirements, and compatible with planned vehicle and mentor certification.
- Motors will be transported by Brendan Williams in their original manufacturer or approved shipping packaging, secured against movement and protected from heat, impact, ignition sources, and unauthorized access. Transportation will comply with applicable DOT and state requirements for the selected motor classification.
- Motors will only be prepared, installed, and used under the direct supervision of Brendan Williams at launches. Motors or propellants will not be modified.
- Motors will remain in the custody of the team mentor Brendan Williams and will be stored according to manufacturer instructions and all applicable federal, state, and local requirements. Motors will be kept in a secured location, separated from ignition sources and incompatible materials, and inaccessible except during supervised activities.
- Ejection charges and other energetic devices will be prepared and handled only under mentor supervision. Charges will remain unarmed and electrically isolated until the appropriate point in the launch procedure.
- Brendan Williams will provide black powder to the team for the ejection charge of the high-power motor.

3.7 Student Briefings and Training

To ensure full-team understanding of safety expectations and practices, we will conduct a mandatory safety briefing chain.

Table 3.7.1: Student Briefings

Time	Subject	Overview	Necessary Personnel
October 3rd	Workspace	Members will be reminded of general workspace safety (access to equipment, PPE, etc.). This includes reminders on basic tools (band saw, and hacksaw, etc.)	Team Educator, all team members
October 3rd	HPR Guidelines / Safety Code	Members will spend time to thoroughly understand the NAR / Tripoli safety codes.	All team members

October 4th	Material Safety	Members will review safety of hazardous materials (handling epoxy, cyanoacrylate, etc.)	All team members
Day Before Each Launch	Review Launch Safety	The Safety Officer will provide a briefing on launch day safety. The team mentor will also provide a rundown on the next day's events.	Team mentor, all team members (including Safety Officer)

3.8 Legal Compliance

Federal Aviation Administration: 14 CFR Part 101, Subpart C

1. We will conduct launches only at approved launch sites, complying with the applicable airspace and FAA restrictions before each flight, ensuring the presence of an adult responsible for the flight.

2. Before each launch, the team Safety Officer will verify the rocket's classification under 14 CFR §101.22 using the motor's certified total impulse and confirming that all applicable FAA requirements are met.

3. For Class 2 or Class 3 launches, we will verify applicable requirements for visibility, cloud conditions, launch times, airport and controlled-airspace restrictions, separation distances, and fire prevention before launches.

4. For launches requiring ATC notification under 14 CFR §101.27, we will provide the required launch information to the appropriate FAA ATC facility within the required 24-hour to 3-day time period.

Bureau of Alcohol, Tobacco, Firearms and Explosives: 27 CFR Part 555

1. We will use only commercially manufactured rocket motors that comply with the applicable requirements of 27 CFR Part 555.

2. Before purchasing or using a motor, our Safety Officer will verify the propellant type and total propellant weight to determine whether it qualifies for the model-rocket exemption under 27 CFR §555.141(a)(10).

3. The team will not manufacture, modify, or alter rocket motors or the rocket propellant. Motors will be purchased, stored, transported, and used under the supervision of the team mentor Brendan Williams in accordance with applicable regulations.

National Fire Protection Association Code 1127:

1. The team will follow the applicable safety and fire-prevention requirements of NFPA 1127 during rocket preparation, motor handling, and launch operations.

2. Appropriate fire-prevention measures will be followed at the launch site, inspecting the launch area before each flight.

Department of Transportation Hazardous Materials Regulations (49 CFR 171–180)

1. The team will purchase rocket motors only from established vendors that classify, package, mark, and ship the motors in accordance with applicable DOT hazardous-material requirements.

2. When transporting rocket motors to launch sites, the team will maintain the motors in their approved packaging and follow all applicable transportation and handling requirements for the motor's DOT classification.

3. Our mentor Brendan Williams will oversee the transportation and handling of rocket motors and verify that the applicable DOT requirements have been satisfied before transportation.

Washington State and Local Requirements:

1. We will verify that each test-launch location is authorized for rocket operations and will obtain permission from the property owner or launch-site operator before conducting launches.

2. We will comply with applicable Washington State requirements governing model and experimental rocketry. Washington's fireworks regulations specifically exclude research and experiments involving rockets and model rocket motors designed for recoverable aero models.

3. Due to the launches being conducted in Pasco, Washington, the team will comply with the applicable Pasco municipal and fire-safety requirements and will coordinate with the Tri-Cities Rocketeers and the launch site's Range Safety Officer before conducting test flights.

4. We will check the current Washington State Department of Natural Resources (DNR) fire-danger restrictions and applicable county/local fire restrictions before each launch. If a restriction prohibits or limits rocket launches, we will postpone or relocate the launch.

Washington State Explosives Act: RCW 70.74

1. Before storing or possessing rocket motors on team facilities, we will consult the Washington State Department of Labor & Industries (L&I) to determine whether the specific motors, quantities, and storage location are subject to Washington explosives licensing or storage requirements.

2. We will obtain any required L&I license or approval before purchasing, possessing, or storing motors that are subject to Washington's explosives requirements and will retain the applicable documentation in the team's safety records. The documentation and requirements will be verified whenever the motor type, propellant quantity, or storage location is changed.

3. We will minimize on-site storage of motors and will not store motors at team or university facilities unless the storage arrangement has been determined to comply with applicable Washington requirements.

Washington State Fire Code — WAC 51-54A (International Fire Code)

1. Before storing or using rocket motors, we will contact the fire code official/AHJ with jurisdiction over the proposed storage location and launch site to determine whether an operational permit or other fire-code approval is required under Chapter 56.

2. The team will comply with applicable local amendments and site-specific requirements identified by the AHJ, including requirements concerning storage, handling, fire prevention, and launch operations.

NASA, NAR, and Tripoli Requirements:

1. We will comply with the current NAR High Power Rocket Safety Code and all applicable Tripoli safety requirements during rocket construction, motor handling, and flight operations.

2. Our mentor Brendan Williams will supervise activities involving rocket motors and other energetic devices and will be present when required by NASA Student Launch requirements.

3. Before each flight, our team's Safety Officer and Range Safety Officer will verify that the launch vehicle, launch site, recovery system, motor, and launch procedures satisfy the applicable NASA Student Launch and NAR/Tripoli requirements.

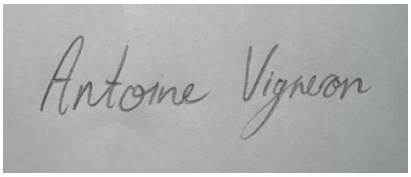
4. We will not launch until the Range Safety Officer has completed the required safety inspection and authorized the flight.

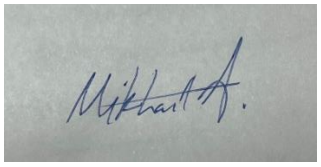
3.9 Safety Agreement

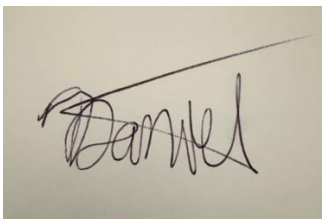
NASA SLI Safety Agreement

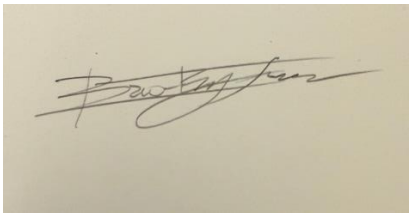
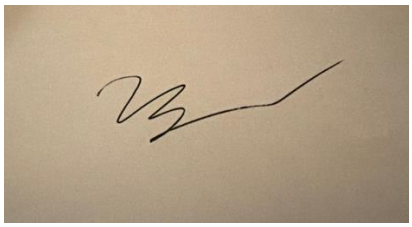
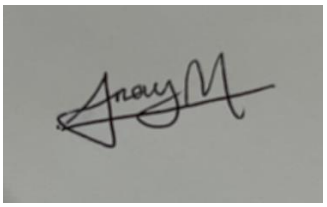
i. Range safety inspections will be conducted on each rocket before it is flown. Each team shall comply with the determination of the safety inspection or may be removed from the program.
ii. The RSO has the final say on all rocket safety issues. Therefore, the RSO has the right to deny the launch of any rocket for safety reasons.
iii. The team mentor is ultimately responsible for the safe flight and recovery of the team's rocket. A team shall not fly a rocket until the mentor has reviewed the design, examined the build, and is satisfied the rocket meets established amateur rocketry design and safety guidelines.
iv. Compliance with NAR High Power Rocket Safety Code.
v. Any team that does not comply with the safety requirements shall not be allowed to launch their rocket.

I understand and will abide to all of these safety regulations:







Date: 9-13-2026

4 Technical Design

4.1 Review Of Launch Vehicle Requirements

This table contains the general requirements as stated in the 2026-27 Student Launch Handbook.

Table 4.1.1: Launch Vehicle Requirements List

	Requirement	Planned Action
2.1	The vehicle shall deliver the payload to an apogee altitude between 3,500 and 5,500 feet above ground level (AGL). Teams flying below 3,000 feet or above 6,000 feet on their competition launch will not be eligible for the Altitude Award	The launch vehicle will be modeled thoroughly in Open Rocket, ensuring that its peak altitude falls safely between 3500 and 5500 feet. The rocket will be constructed as designed.
2.2	The launch vehicle and payload SHALL be capable of remaining in launch-ready configuration on the pad for a minimum of 3 hours without losing the functionality of any critical on-board components, although the capability to withstand longer delays is highly encouraged	Electronic equipment will be powered with sufficiently large batteries. This will be tested over multiple occasions before flight.
2.3	Teams shall declare their target altitude goal at the CDR milestone. The declared target altitude shall be used to determine the team's altitude score.	The target altitude will be included in the CDR.
2.4	The launch vehicle shall be designed to be recoverable and reusable. Reusable is defined as being able to	The vehicle will be designed and constructed with adequate robustness. Numerous tests will be conducted throughout the season to ensure our design adheres to

	launch again on the same day without repairs or modifications.	standards surround launch loads. All parts of the rocket that can be made reusable will be made reusable.
2.5	The launch vehicle shall have a maximum of four (4) independent sections. An independent section is defined as a section that is either tethered to the main vehicle or is recovered separately from the main vehicle using its own parachute.	The rocket will split into 3 independent sections, separated by the parachutes. The payload is captive and the rocket will be recovered in one tethered piece.
2.5.1	Coupler/airframe shoulders which are located at in-flight separation points shall be at least two airframe diameters in length. (one body diameter of surface contact with each airframe section).	All connection points (av bay coupler) will have at least 1 body diameter of connection length with each tube.
2.5.2	Coupler/airframe shoulders which are located at non-in-flight separation points shall be at least 1.5 airframe diameters in length. (0.75 body diameter of surface contact with each airframe section.)	The rocket will be manufactured with non-inflight-separation points of at least 1.5 airframe diameters in length.
2.5.3	Nosecone shoulders shall be at least ½ body diameter in length.	The nosecone used will have a shoulder length of at least ½ the airframe diameter.
2.6	The launch vehicle shall be capable of being launched by a standard 12-volt direct current firing system. The firing system shall be provided by the NASA-designated launch services provider.	The chosen igniter for the motor will be capable of being launched with the standard 12 volt DC firing system.
2.6.1	Each team SHALL use commercially available ematches or igniters. Hand-dipped igniters SHALL not be permitted.	Commercially available e-matches will be purchased and used for pyrotechnics.
2.7	The launch vehicle shall use a commercially available solid motor propulsion system using ammonium perchlorate composite propellant (APCP) which is approved and	A commercially available motor that uses APCP and is approved by the NAR and TRA will be chosen for the launch vehicle.

	certified by the National Association of Rocketry (NAR), Tripoli Rocketry Association (TRA), and/or the Canadian Association of Rocketry (CAR).	
2.7.1	If a team is choosing its competition motor from the NASA-supported list up above, the team shall declare at the Preliminary Design Review (PDR) the total number of motors they intend to purchase for the entire season. This number shall reflect the quantity the team intends to buy, not any motors currently in their possession.	If our team decides to use a motor included in the NASA supported list, we shall declare the number we intend to purchase for the season at the PDR.
2.8	The launch vehicle shall be limited to a single motor propulsion system.	The launch vehicle will be designed with a propulsion system limited to 1 motor
2.11	The total impulse provided by a High School or Middle School launch vehicle shall not exceed 2,560 Newton-seconds (K-class).	The chosen motor for the design will be limited to the K class.
2.12.1	The minimum factor of safety (Burst or Ultimate pressure versus Max Expected Operating Pressure) will be 4:1 with supporting design documentation included in all milestone reviews.	N/A. The vehicle will not include any pressurized tanks or compressed gas storage.
2.12.2	Each pressure vessel shall include a pressure relief valve that sees the full pressure of the tank and is capable of withstanding the maximum pressure and flow rate of the tank.	N/A. The vehicle will not include any pressurized tanks or compressed gas storage.
2.12.3	The full pedigree of the tank shall be described, including the application for which the tank was designed and the history of the tank. This will include the number of pressure cycles put on the tank, the dates of pressurization/depressurization, and	N/A. The vehicle will not include any pressurized tanks or compressed gas storage.

	the name of the person or entity administering each pressure event.	
2.13	The launch vehicle shall have a minimum static stability margin of 2.0 while sitting on the pad.	The rocket will be designed thoroughly and accurately in open rocket with a confirmed stability margin of 2.0, and construction will adhere completely to the design.
2.14	The launch vehicle shall have a minimum thrust to weight ratio of 5.0:1.0	The rocket will be designed with a maximum weight of 5 times less than the thrust of the chosen motor.
2.15	Any structural protuberance on the rocket shall be located aft of the burnout center of gravity. Camera housings will be exempted, provided the team can show that the housing(s) causes minimal aerodynamic effect on the rocket's stability	No structural protuberance on the rocket design will be located ahead of the burnout CoG. If the team decides to use a camera and housing, it will be confirmed that the housing causes minimal aerodynamic effects.
2.16	The launch vehicle shall accelerate to a minimum velocity of 52 fps at rail exit.	The launch vehicle will be testing in simulation before flight, ensuring that it reaches a minimum velocity of 52 feet per second upon rail exit.
2.17	Subscale rockets are required to use a minimum motor impulse class of E (mid-power motor)	The design of the subscale rocket will involve a motor of class E or above.
2.18	The subscale rocket shall not exceed 75% of the dimensions (length and diameter) of your designed full-scale rocket. For example, if your full-scale rocket is a 4" diameter, 100" length rocket, your subscale shall not exceed 3" diameter and 75" in length.	The subscale rocket will be designed to remain under 75% of the dimensions of the full scale rocket.
2.19-20	All teams shall complete demonstration flights as outlined below: Vehicle Demonstration Flight—The purpose of the Vehicle Demonstration Flight is to validate the launch vehicle's stability, structural integrity, recovery systems, and the team's ability to prepare the launch vehicle for flight.	A demonstration flight will be conducted in order to demonstrate the vehicles stability, structural integrity, and recovery system. The flight will aim to demonstrate proper functionality of all launch vehicle's hardware components.

	A successful flight is defined as a launch in which all hardware is functioning properly (drogue chute at apogee, main chute at the intended lower altitude, functioning tracking devices, etc.).	
2.21	All Lithium Polymer batteries shall be sufficiently protected from impact with the ground and will be brightly colored, clearly marked as a fire hazard, and easily distinguishable from other payload hardware.	Any and all lithium polymer batteries will have sufficient protective housing with brightly colored marking to signify the fire hazard
2.22	Vehicle Prohibitions	The launch vehicle design and construction will adhere to all given vehicle prohibitions. This includes: • The launch vehicle will not use forward firing motors • The launch vehicle will not utilize motors that expel titanium sponges • The launch vehicle will not utilize hybrid motors • The launch vehicle will not utilize a cluster of motors • The launch vehicle will not utilize friction fit motors • The launch vehicle will not exceed Mach 1 at any point in its flight • Vehicle ballast will not exceed 10% of the total unballasted weight of the rocket, as it would sit on the pad. • Transmissions from on-board transmitters shall not exceed 250 mW of power.

4.2 Launch Vehicle Design

4.2.1 General Airframe Info

The preliminary launch vehicle design is 95.2” long and 5.5” in diameter. The airframe is composed of blue tube, a proprietary creation of Always Ready Rocketry, very similar to vulcanized cellulose fiber tubes. The design also features a 3d printed nosecone paired with 3 ¼” thick non-canted plywood fins. It weighs 22 pounds on the launch pad, with a static stability margin of 2.47. It is designed for a 54mm K-class motor.



4.2.2 Separations

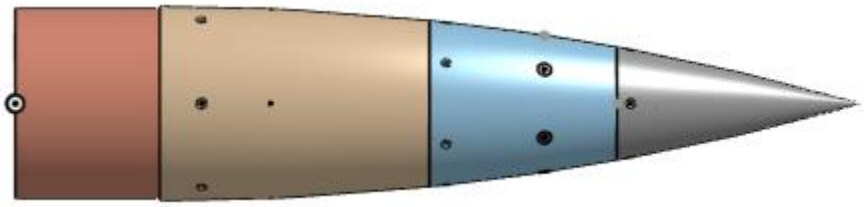
The rocket features two primary separation points; one just aft of the avionics bay, and one just ahead of the main parachute bay



The blue pin marks the first separation point - the drogue chute. This separation occurs at apogee. The red pin marks the second separation point – the main chute. This deploys around 600 feet A.

4.2.3 Nose Cone

The nose cone of the rocket will be 3d printed in Polyethylene Terephthalate Glycol, a tough alternative to polylactic acid typically capable of heat deflection around 174 degrees Fahrenheit. Computational fluid dynamics will be performed during the season to create an objective estimate of the stagnation force and stagnation temperature on the nose cone. This will allow us to verify the integrity of PETG as the nosecone material. In the meantime, a more preliminary analysis to narrow in on PETG was also conducted. This is covered in the **materials** segment. The print is created in 4 separate parts, and in this preliminary design, these parts are locked tight together by around 16-18 flush-mounted screws. This quantity is flexible and can be adjusted as needed. The nosecone will also be tested and iterated during the season through things like static fires to ensure that it withstands the ejection and aerodynamic-related force loads expected in the flights. The nosecone is designed with a 4.125” shoulder, meeting the 0.75*airframe requirement. This design choice reflects the desire to maximize space in the main parachute bay for packing the recovery system. The nose cone is also designed with a flight computer and various pressure ports. More details regarding this interior design are included in the **payload** segment of this proposal.



Preliminary computer-aided design of rocket nosecone, made in and publicly available on Onshape. After initial tests, if a smoother shape is desired, the initial design may be altered to allow for longitudinal printing.

4.2.4 Main Chute Bay

The main chute bay is designed as a 25” by 5.5” blue tube airframe. Attached via 4-8 nuts and bolts to the rocket avionics bay, this section of the rocket will house the main parachute and its harness. It will remain permanently fixed to the avionics bay throughout the flight, while its connection to the nose cone shoulder will be blown apart by pyrotechnics. This tube will also host a Kevlar section of shock cord and a reusable, 18” apogee components fire blanket to protect the recovery equipment from any burning by the pyrotechnics.

4.2.5 Avionics Bay

The avionics bay of the rocket features a 14-inch blue tube coupler paired with a 2-inch switch-band centered on the coupler. This switch band will have static vent ports, a switch port, and 6 isolated pressure taps for the payload experiment. Each side of the avionics bay will have one coupler bulkhead as well as one tube bulkhead, tied together by two bolted M10 rods. These two bulk heads will also be attached to stainless steel U-Bolts, tying in the harness. They will have wiring ports for both the primary and back up pyrotechnical systems. They will each be of 0.25” thick plywood. This avionics bay connects directly to the drogue chute, pulling it out of the fin can upon separation. It will host two independent altimeters, a GPS tracker, ejection-charge related hardware, telemetry hardware, and a custom flight computer. Further details regarding this flight computer are included in the **payload** segment.

4.2.6 Fin Can Tube

The fin can tube is designed as a 48” by 5.5” blue tube airframe. Designed to be separated from the avionics bay, this tube hosts the drogue parachute and its harness, as well as a Kevlar section and reusable 18” fire blanket. The fin can also host the 27.5” by 54 mm blue tube motor mount. This mount is centered via 4 0.25” thick plywood centering rings; one mounted at the back of the airframe, one mounted just forward of the fin slots, and two stacked towards the front of the motor mount tube. A 54mm screw on metal motor retainer is used to secure the motor in place. This retainer is the Aeropack 54mm L Retainer, RA54L, secured to the motor mount via epoxy. More details regarding motor choice are included in the **motor** segment. To attach the harness to this section of the airframe, the same stainless steel U bolt will be attached / bolted to the double centering ring on the forward section of the motor mount. To improve structural integrity of this setup, extra epoxy will be applied around these rings. More details regarding this process can be found in the **construction** section Furthermore, tests will be conducted post-construction to ensure the design can tolerate ejection-related loads from the harness.

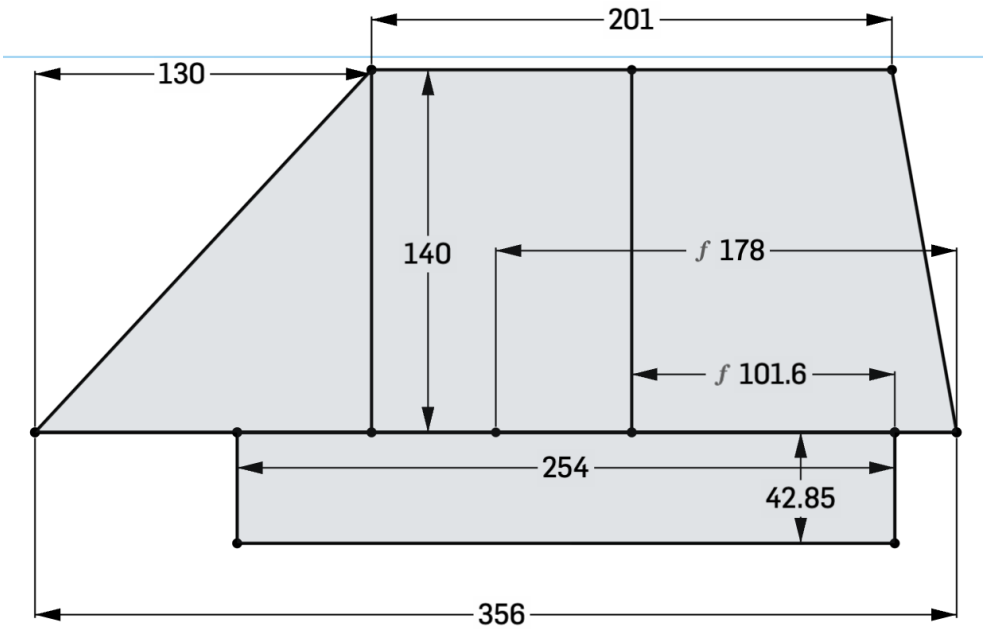
To create a strong airframe, the fins will be connected to the motor mount via fin tabs. To allow these fin tabs to enter the rocket body, slots will be cut at the appropriate section of the fin can tube. Prioritizing proper alignment, these slots will be cut using a custom designed and 3d printed stencil. This stencil will slide flush onto the aft part of the tube, taped in place as the fin slots are cut with a Dremel-like rotary tool. In this preliminary design, these fin tabs are 10 inches in length and 0.25 inches in width.

4.2.7 Fins

The preliminary design consists of trapezoidal fins. This plywood fin shape was chosen due to the following factors:

- Ease of manufacturing
    - Entirely composed of straight lines, allowing for clean cuts. This can apply if we switch to using fiberglass fins or lose access to the laser cutter.
- High efficiency
    - Approaches the aerodynamic efficiency of elliptical fins whilst still maintaining large surface area for adequate corrective force and stability.
- Maintains robustness and mechanical strength

Figure 4.2.1: A sketch of the preliminary fins. Dimensions are in millimeters.



The initial fin dimensions:

- Height: 140mm
- Tip Chord: 201mm
- Sweep Length: 130mm
- Root Chord: 356mm

The preliminary design of these fins is not optimized. Rather, this preliminary design's dimensions were chosen due to general rocket-wide trends, ease of manufacturing, and the maintenance of adequate static stability. The trapezoidal shape maintains easy manufacturing and construction as it is comprised entirely of straight lines. These fins will be epoxied to the central blue tube 54mm motor mount via the fin tabs.

4.2.8 Fillets

Fillets made from JB SteelWeld will be utilized to improve the structural integrity of the fin can. More information regarding this process can be found in section 4.4.

4.2.9 Air-Foiling Fins

Though air-foiling can significantly improve the aerodynamics of the launch vehicle's fins, the preliminary design does not utilize this technique due to the complexity of uniform air-foiling. This thus reduces the chance of unintended asymmetries in the rocket design.

4.3 Materials (choices)

4.3.1 Nose Cone – PETG

Though a Mach ~ 0.6 rocket generally does not require exceptional nose cone strength, any deformations in the nosecone profile could significantly impact our experiment. This is especially true considering the presence of heat brought on by ram airflow on the nosecone tip. 3d printed materials often weaken or begin to deform as temperatures increase. Thus, choosing an ideal filament material is crucial for the project. As stated above, a deeper, physics-based or CFD analysis would have to be conducted to prove the adequacy of the chosen filament type. However, a general comparison between commercially available filaments can still be made:

Table 4.3.1: 3D Printer Filament Comparison

Filament Type	Tensile Strength	Heat Deflection Temp	Weight	Impact Strength	Enclosure	Bambu Labs A1
PLA	50-60 MPa	122 F	1.24 g/cm	Low	No	Yes
PETG	45-55 MPa	172 F	1.27 g/cm	Moderate to High	No	Yes
ABS	35-45 MPa	200+ F	1.04 g/cm	High	Yes	No
PLA+	55-65 MPa	127 F	1.24 g/cm	Moderate	No	Yes
ASA	35-52 MPa	200+F	1.06 g/cm	High	Recommended	No

Our team does not have access to proper, enclosed 3d printers. And though the filaments associated with the Bambu Lab A1 printers, which we do have, are heavier and less temperature resistant compared to ASA and ABS, they are all we currently have access to. Among the remaining 3 filament types, though PETG features the weakest tensile strength, its ability to absorb impact and deflect heat is superior to that of PLA or PLA+.

4.3.2 Tube – Blue Tube

There are a variety of known tube materials for L2-scale rockets. For our design, affordability, workability, and strength were prioritized. To decide on using blue tube 2.0, a general review of a multitude of options was conducted:
Cardboard / Paper Composites (thick wall):
This is easily the most affordable tubing option, works fresh out of the box, and is affordable + easy to cut. However, it is also the weakest material and can often dent, especially under transverse impact.

Phenolic:
Hard, dense, heat cured composite cylinders made by wrapping a base material around a mandrel under heat with a phenolic resin binder. These tubes are extremely brittle, however. This makes them prone to zippering and makes cutting harder. Often, preventing cracking involves adding a layer of fiberglass- a messy process.

Fiber Glass:
Very high compressive strength but is quite heavy and involves complexities with management. It is also quite brittle and very much expensive.

Blue Tube:
As proprietary creation of Always Ready Rocketry very similar to sorts of vulcanized fiber, Blue tube is generally tougher than phenolic tubes, and does not feature the brittleness that often characterizes the latter. These benefits come at the expense of mass- blue tube weighs about 20% more than phenolic.

Carbon Fiber:
The most extreme material, carbon fiber features incredibly high strength to weight ratios. However, dealing with carbon fiber is incredibly dangerous, and its dust leads to permanent lung damage or severe skin irritation. Carbon fiber is also the most expensive material on this list.

Considering these options, Blue tube provided the best balance as a relatively lightweight, affordable, and workable material that was still capable of high load bearing or direct impact.

4.3.3 Fins: Plywood

Material choices for the Launch Vehicle's fins are more limited. Generally, it comes down to three primary options:

Carbon Fiber: Very strong, often very expensive and difficult to work with.

Aircraft Plywood: Lightweight, porous, generally weaker than other options at this level.

G10 Fiberglass: Similar in weight to aircraft plywood if 0.125" thickness is used. Though stronger, fiberglass is also much harder to work with, requiring much heavier PPE, and is more challenging to use with adhesives. Fiberglass does have the benefit of not warping under moisture, but proper sealing / painting of aircraft plywood could make up for this.

Thus, Aircraft Plywood was chosen. Further tests will be conducted prior to construction to confirm this material's integrity for our rocket. In the case that it is not, fiberglass fins of ½ the thickness would fair as a solid substitute, retaining the same mass as the plywood fins.

4.4 Construction

4.4.1 Blue Tube

Blue tube will be cut transversely using a hacksaw and / or bandsaw, depending on availability of the machine. As mentioned previously, fin slots will be cut with a Goxawee rotary tool (similar to a Dremel), and a 3d printed stencil. This stencil will be connected flush to the tube, allowing for precise cutting of the TTW fin slots. To prevent dulling or shattering of the tool, a Fiberglass reinforced cut-off wheel will be used. This should enable smooth cutting of the blue tube.

4.4.2 Fins

The aircraft plywood fins will be cut by a laser cutter. Flat, commercially available ¼ inch aircraft plywood plates will be purchased and cut to the dimensions of the above fin design. The laser cutter will be operated safely, with both adult and safety lead supervision. These supervisors will also double check the tightness of the laser alignment screw.

4.4.3 Motor Mount

Two centering rings will have holes drilled into them to mount the U-bolt. They will be epoxied together, and the U-bolt will be screwed on. The nut will then be tightened and locked in place with epoxy. The motor mount will be lightly sanded down at points of adhesion, and a slow-cure epoxy like BSI 30 minute epoxy will be used to attach all 4 centering rings to the tube. Meanwhile, the Aeropack 54mm – L motor retainer will be glued onto the motor mount using JB SteelWeld epoxy. This epoxy features greater resistance to heat and therefore, despite being heavier than clear epoxy, is more suited for the task of maintaining adhesion between the motor mount and the motor retainer.

To glue this motor mount into the Blue Tube Fin can, a coarse sandpaper will be used to scuff the key points of adhesion around the centering rings and inside the blue tube. A dowel will then be flagged to the locations of adhesion within the fin can. A thick, slow cure epoxy (ex. West system 105-206 or BSI slow cure) will be used for the adhesion. After all four rings are glued in place, extra epoxy will be applied around the front of the forward-most centering ring and around the back of the aft most centering ring.

4.4.4 Glueing Fins

The same structural epoxy will be used to glue the fins in through the fin tabs. Before adhesion, coarse sandpaper will be used on the bottom side of the fins to improve joint quality. A custom laser-cut cardboard fin jig may be used to ensure that each fin is glued on straight.

4.4.5 Fillets

The fillets of the rocket will be done using JBweld SteelWeld original epoxy. A small layer will be evenly spread along the edge of connection between the fins and the fin can tube. A popsicle stick or 3d printed piece may be used to smooth this fillet out evenly.

4.4.6 Couplers

Will be cut longitudinally with a hack saw / bandsaw depending on machine availability. Slow cure epoxy will be used to glue the coupler within the external blue tube after both surfaces are scuffed with a coarse grain sandpaper.

4.4.7 Drilling

All holes will be drilled using a modern, electric hand drill with standard commercially available bits. Alternatively, Ryobi Forstner bits may be used as needed.

4.5 Projected Altitude & Choosing Motor

The launch vehicle will target a peak altitude of 4300 feet AGL. This altitude was calculated using OpenRocket's numerical physics solver with a variety of high-total impulse K class motors. To decide on a target altitude, a consensus on 1-2 similar chosen motors. Choosing these motors would require a brief scan over a variety of options.

In the chart below, availability is measured via the amount of sites on the first two google search pages with this motor in stock.
The velocity off rod is calculated in m/s off of a 12-foot rod. The required rail exit speed of 52fps translates to about 15.846 meters per second.
Though cost varies, the cost in the table reflects an average, no shipping or tax base cost.

Table 4.5.1: Motor Options

Motor	Velocity off rod	Apogee	Cost	Availability	Needs Casing
CTI K815	24.9	4390	305-400	1	Y
CTI K300	18	3766	300	0	Y
Aerotech K700	27.1	2394	260	2	Y
Aerotech K1275	34.3	1753	260-400	3	Y

CTI K260	16.5	3425	250	0	Y
Aerotech K480	24	2755	260	2	Y
Aerotech K455	31.3	4523	300-330	2-3	N
Aerotech K250	15.1	4083	388	1	N
CTI K750	25.9	4335	350	0	Y
CTI K780	25.5	4014	240	0	Y

Out of these options, the best balance between availability, peak altitude, and cost was the Aerotech K455, our preliminary motor choice.

Given this motor choice, accounting for extra mass from extra glue, extra mass from small screws on the rocket that were not accounted for, and altitude lost from weathercocking, the target apogee of 4300 feet AGL was chosen.

The use of other solvers or more thorough / accurate simulation modeling may enable a more accurate prediction later on.

4.6 Recovery

As stated above, the launch vehicle design features a dual deployment recovery system. The drogue chute deploys at apogee, while the main parachute will deploy at around 600 ft. The avionics bay will consistent of two independent altimeters firing two rounds of charges per separation; one main, and one back up.

Table 4.6.1: Requirements Table

3.1	The full-scale launch vehicle shall stage the deployment of its recovery devices, where a drogue parachute is deployed at apogee, and a main parachute is deployed at a lower altitude. Tumble or streamer recovery from apogee to main parachute deployment is also permissible, provided that kinetic energy during drogue stage descent is reasonable, as deemed by the RSO.	The launch vehicle will be designed and constructed with the use of a small drogue parachute that deploys at apogee, followed by a main parachute at 600 feet.
3.1.1	The main parachute shall be deployed no lower than 500 feet.	The launch vehicle and its electronics will be designed / programed in a way so that the main charge will fire at 600 feet, the back up charge will fire at 525 feet.

3.1.2	The apogee event shall contain a delay of no more than 2 seconds.	Adequately large static pressure vent ports will be used to prevent delay. Static fires will be performed to ensure smooth separation.
3.1.3	Motor ejection is not a permissible form of primary or secondary deployment	The avionics bay will be designed to control both deployments. The motor will have no ejection charge.
3.2	Each independent section of the launch vehicle shall have a maximum kinetic energy of 75ft - lbf at landing	General physics estimations will be performed to ensure that the kinetic energy upon landing (with the main parachute deployed) falls well under 100 joules.
3.3	The recovery system shall contain redundant, commercially available barometric altimeters that are specifically designed for initiation of rocketry recovery events. The term “altimeters” includes both simple altimeters and more sophisticated flight computers.	Two independent commercially available altimeters will be used for the two deployments.
3.4	Each altimeter shall have a dedicated power supply, and all recovery electronics shall be powered by commercially available batteries.	The recovery system will be designed so that each independent altimeter has its own power supply. It will also be designed so that all electronics may be powered by commercially available batteries.
3.5	Each altimeter shall be armed by a dedicated mechanical arming switch that is accessible from the exterior of the rocket airframe when the rocket is in the launch configuration on the launch pad.	The recovery system’s avionics bay will be armed via a pin switch accessible from the exterior of the rocket whilst it is on the launch pad.
3.6	Each arming switch shall be capable of being locked in the ON position for launch (i.e., cannot be disarmed due to flight forces).	The recovery system will be designed so that flight forces cannot impact the arming switch, and it will be locked in the ON position.
3.7	The recovery system, GPS and altimeters, and electrical circuits shall be completely independent of any payload electrical circuits.	The flight computer and other electronics associated with the payload experiment will be entirely independent of recovery system electronics per the design of the recovery and payload systems.
3.8	Removable shear pins shall be used for both the main parachute compartment and the drogue parachute compartment.	The recovery system will be designed with shear pins that snap cleanly during both separations. Static fires will be performed to ensure that this system is functional.

3.9	Bent eyebolts shall not be permitted in the recovery subsystem.	Bent eyebolts will not be used on the launch vehicle whatsoever.
3.10	The recovery area shall be limited to a 2,500 ft. radius from the launch pads.	Calculations will be performed during the season to estimate the drift of the rocket with varying parachute sizes. These calculations will fuel the final recovery system design, ensuring that the rocket does not drift outside of the 2500 foot radius from the launch pads.
3.11	Descent time of the launch vehicle shall be limited to 100 seconds (apogee to touch down).	Calculations will be performed during the season to estimate the descent time of the rocket with varying parachute sizes. These calculations will fuel the final recovery system design, ensuring that the rocket does not descend for a time period longer than 100 seconds.
3.12	Electronic GPS Tracking device(s) shall be installed in the launch vehicle and will transmit the position of the tethered vehicle and any independent section(s) to a ground receiver.	The avionics bay will be designed to host a GPS tracking device transmitting the position of the vehicle (fully tethered) to a ground receiver.
3.13.1	The recovery system altimeters shall be physically located in a separate compartment within the vehicle from any other radio frequency transmitting device and/or magnetic wave producing device.	The avionics bay will be designed to host the recovery system altimeters in a compartment separate from any RF transmitting / magnetic wave producing device.
3.13.2	The recovery system electronics shall be shielded from all on-board transmitting devices to avoid inadvertent excitation of the recovery system electronics.	Any and all recovery system-related electronics will be designed to be shielded from any and all onboard transmitting devices.
3.13.3	The recovery system electronics shall be shielded from all on-board devices which may generate magnetic waves (such as generators, solenoid valves, and Tesla coils) to avoid inadvertent excitation of the recovery system.	Any and all recovery system-related electronics will be designed to be shielded from any and all onboard devices that may generate magnetic waves.
3.13.4	The recovery system electronics shall be shielded from any other on-board devices which may adversely affect the	Any and all recovery system-related electrons will be designed to be shielded from any and all onboard devices that may adversely affect the proper operation of the recovery system electronics.

	proper operation of the recovery system electronics	

4.6.1 Parachutes - Design, Sizing, and Calculations

One parachute will be utilized for each separation event. These parachutes will be deployed via the ignition of black powder charges positioned on both sides of the avionics bay. These charges create pressure within the parachute bays that strips away the nylon shear pins and separates the components. The design and sizing of these parachutes will largely rely on calculations performed regarding descent times and drift distance of the launch vehicle, as well as reducing kinetic energy of the rocket upon landing. Parachutes will be purchased from external vendors. Although an in-depth CFD simulation would have to be performed to accurately predict descent times, velocities, and drift, a preliminary, worst-case-scenario analysis can still be performed:

Cd of parachute + payload:

$$C_d = \frac{2 \cdot W}{\rho \cdot V^2 \cdot A}$$

And thus:

$$V = \sqrt{\frac{2 \cdot m \cdot g}{\rho \cdot C_d \cdot \pi \cdot r^2}}$$

Or for two parachutes:

$$V = \sqrt{\frac{2 \cdot m \cdot g}{\rho \cdot \pi (C_{d1} \cdot r_1^2 + C_{d2} \cdot r_2^2)}}$$

Air density at sea level is 1.225 kg/m^3. Air density at 5000 ft. is around 1.056 kg/m^3. These values average out to 1.14 kg/m^3
Burnout mass: 9 kg
g = 9.81

Using the Rocketman Parachutes as a baseline for parachute sizes and Cd:
The pro experimental drogue parachutes have a drag coefficient of 0.97
The High-Performance parachutes have a drag coefficient of 2.2

Given that the maximum kinetic energy on descent is 75ft-lbf or around 100J, with a 9kg rocket,

$$\frac{mv^2}{2} = K_e$$

and

$$v = \sqrt{\frac{200}{9}} = 4.71$$

Gives 4.71 meters per second. Assuming the parachute deploys at 550 feet, this secondary descent phase will take at least 35 seconds, realistically 40 (to be closer to the window). This leaves 40-50 seconds (10 seconds of extra room) for the drogue to descend over 4300-550 = 3750 feet. This results in a target drogue descent rate of around 80 feet per second.

A descent rate of 24.834 m/s, mass of 9kg, a coefficient of drag of 0.97, and air density of 1.14 kg/m^3,

$$24.834 = \sqrt{\frac{176.58}{1.14 * 0.97 * A}}$$

and thus A = 0.2589 meters ^2

This translates to around a 20inch diameter parachute, or around 2.787 square feet. The closest Rocketman Parachute size to this is the 2ft standard parachute, with a surface area of 2.6 square feet. This results in a recalculation of drogue descent rate to around 84.8 ft/s.

Given the minimum descent speed of 4.71 meters per second after the second separation, we can find a minimum main parachute size.

$$4.71 = \sqrt{\frac{2 \cdot 9 \cdot 9.81}{1.14 \cdot (0.97 \cdot 0.2589 + 2.2 \cdot A)}}$$

Thus, the minimum area of the main parachute is calculated to be 32.933 feet squared. The 60 inch High Performance CD has a canopy area of 33.86, while the 72 inch has a canopy area of 48.76. The 72 inch chute was chosen to ensure adequately low landing kinetic energy.

To calculate drift, total descent time must be calculated.
Recalculating main chute descent time gives 12.7953 feet per second. This results in a total descent time of 87 seconds.
As an overestimation / worst case scenario, the drift of the rocket = descent time * wind speed.
This means that the rocket would only drift out of the designated range under 30+ mph wind speeds across all altitudes.

4.6.2 Avionics Bay Design – Recovery

The avionics bay, located between the main parachute bay and fin can tube / drogue parachute bay, will consistent of a special, two-part compartment designated specifically for the recovery system electronics. This compartment will be completely isolated from payload-related electronics. One part of this compartment will contain GPS transmission. The other isolated part will contain the two independent altimeters, two independent batteries to power the altimeters, and two pull pin switches for each altimeter, accessible from the exterior of the launch vehicle.

As stated above, the avionics bay will feature 2 stacked bulk heads on either side skewered by two M10 Rods. Each side will consist of a U bolt, which connects to the harness, tethering in the parachutes, fin can, and nose cone. To prevent the drogue chute from getting stuck in the fin can tube, it will be tethered directly to the avionics bay U bolt. This means that as the tubes separate, the avionics bay pulls the drogue chute out from the fin can allow for smooth deployment. The bulkheads will host the ejection charge canisters, and consistent of ports for E-match / igniter related wiring to route the altimeters to their respective charge canisters. It is also crucial to ensure that each match rests fully within the charge.

4.6.3 Recovery Electronics

The recovery system will utilize the Eggfinder Mini Transmitter as the primary GPS module. This module is a mere 0.7” wide and 3.25” long (including the antenna). This is a budget friendly GPS transmitter capable of transmitting over around 6 miles of distance- far more than expected from this launch vehicle. It will be assembled using standard soldering equipment available to the team.

The altimeters will be PerfectFlite StratoLoggerCFs (compact footprint). These altimeters can measure up to 100,000 feet with 0.1% accuracy. They will each be connected via e match to the ejection charge. Furthermore, they will be tested extensively through ground tests and the subscale flight. The altimeter compartment will have direct access to pressure vent ports to ensure fidelity of measurements.

These 3 isolated circuits will be powered by 3 separated batteries. The chosen battery is a 7.4Volt 2-series 300 mAh Lithium Polymer battery. These batteries should provide more than enough current to power the e-matches.

Each circuit will also include a pull pin switch that can be accessed externally on the rocket.

4.6.4 Ejection Charges

As per typical practice, 4F (FFFFg) black powder will be held in the 4 ejection canisters. Each separation event will have both a main and a backup charge ignition. These canisters are purchased from apogee components and are sealed off with tape. During the season, deeper analysis into the necessary quantity of black powder within each canister will be performed. Furthermore, static fires with shear pins will be conducted to ensure smooth separation.

4.7 Payload

The proposed launch vehicle utilizes a fully captive payload using onboard sensors to answer the following research question:

Can a neural network trained off both CFD and real-world flight data work as a surrogate to accurately predict surface pressure at a variety of locations around the rocket nosecone and avionics bay during high-speed flight?

4.7.1 Overview of Proposed Experiment

The goal of the experiment is to prove the efficacy of a neural network trained on both CFD and real-world data as a surrogate to predict and estimate aerodynamics around the rocket. Given the nature of this project as a proof of concept, only two regions were chosen for this aerodynamic prediction. The first region exists 9 inches aft of the tip of the nosecone. The second region exists around the exposed section of the avionics bay. Each region features 6 circumferentially distributed pressure taps.

To perform this experiment, CFD will first be performed over the nosecone, main chute bay, and avionics bay. These runs will occur over varying velocities, roll rates, and angles of attack. Then, a neural network will be trained off this data, aiming to predict the surface pressure at 6 locations. This neural network will also be trained on the demonstration flight. If needed, more full-scale or sub-scale flights may be performed during the season.

4.7.2 Experimental Procedure

Question: What is the efficacy of correcting a CFD surrogate model with a correction model trained off limited real-world data?

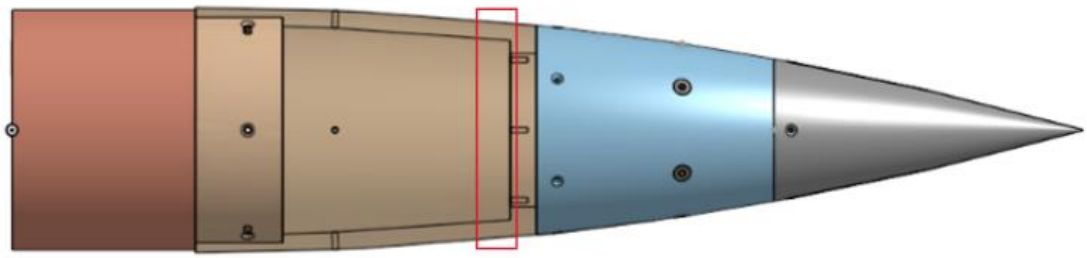
The CFD surrogate model will be evaluated on its accuracy to real world observed data. During the experiment flight, the model will be given the decided upon set of pressure data along with all other observation fields it is trained on. This serves as a control for the correction model while also providing insight into CFD, while the withheld data that is predicted serves as a ground truth.

The correction model, applied to a new situation will be given the surrogate model's predictions, and the results of its corrected predictions will be compared against the base CFD surrogate model's error.

4.7.3 Design Within the Launch Vehicle

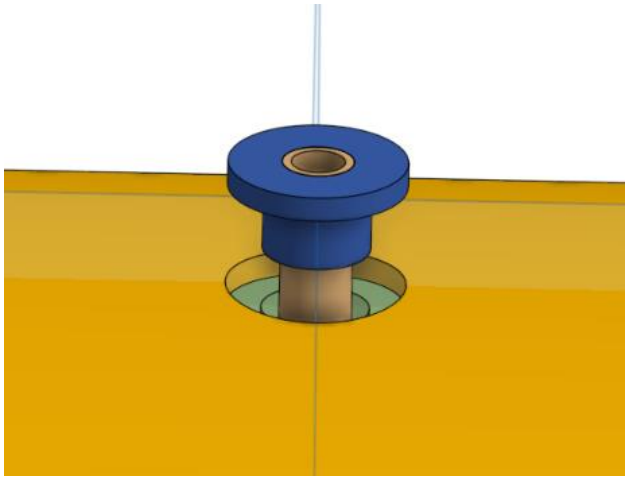
Two independent, fully custom flight computers will be hosted inside of the Launch Vehicle. One will be located inside of the nosecone, while the other will be located inside of the avionics bay.

Figure 4.7.1: The nosecone flight computer will be mounted in the red box area, screwed in place



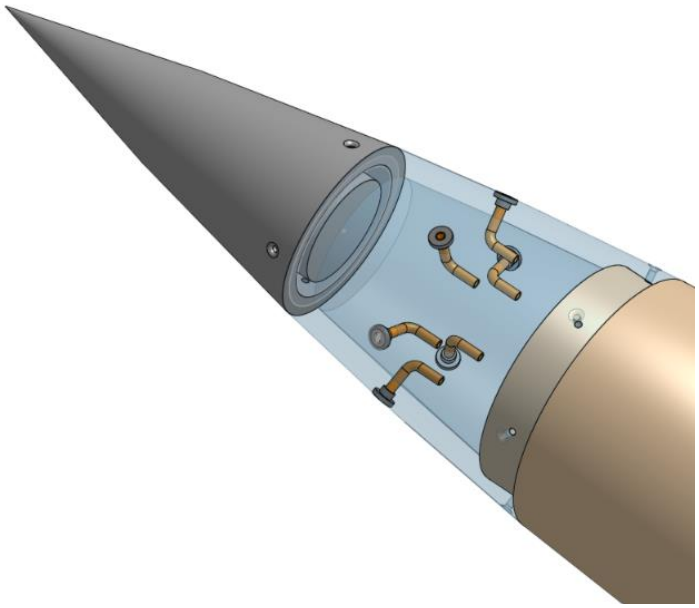
The forward side of the nosecone flight computer will connect to the 6 flush pressure taps located on the blue segment of the nosecone.

Figure 4.7.2: Zoomed in profile of pressure taps. The yellow surface is the switch band, with the green underneath being the tube coupler. The flush 3d printed mount is blue, and the tan tube is a copper capillary.



Both the avionics bay and the nosecone feature 6 circumferentially spaced pressure taps. These taps feature a flush, 3d printed mount that centers and locks a copper capillary tube. This capillary tube, dimensions ID 3mm OD 4mm, sticks out within the bay. Then, a silicone capillary tube, dimensions ID 4mm OD 5mm, is slid over this copper capillary, as well as the individual pressure sensors on the flight computer.

Figure 4.7.3: A CAD model showing the interior pressure tubing within the Launch Vehicle's Nose Cone



4.7.4 Computational Fluid Dynamics

Computational fluid dynamics will be used to generate a pressure dataset that will initially train the neural network. Each simulation will target a static point in the rocket's flight, and the corresponding flight state will be used to construct the simulation. The flight state will be comprised of parameters like velocity and angle of attack, for example.

To ensure an accurate simulation, we focus on multiple aspects of the simulation to model rocket flight.

- Overall:
    - CFD Pipeline (pressure probes)
    - Coordinate system and rocket geometry
    - Bounding box
    - Inlet/outlet
    - The rocket is stationary while air flows around it, or the rocket is modeled as moving
    - How many simulation cases?
    - What parameters will be adjusted (for now)
        - Velocity: show flight profile graph from openrocket simulation
- Iteration convergence
- Meshing
    - Base mesh
    - Near-field mesh refinement
- Turbulence model
    - Spalart-Allamaras
        - This turbulence model features low computational overhead, as well as fast convergence for attached flows. For our rocket, we do not expect prolonged high angle of attack durations, as the stability margin is relatively high.
    - Shear Stress Transport
        - This turbulence model is generally adopted for most aerospace applications.
- Slip conditions/boundary layer
- Atmospheric conditions
- Validation against actually rocket and pressure sensors
- Pressure probes in the simulation
- Limitations:
    - Unmodeled Vehicle Motion
    - Uncertainty in certain atmospheric conditions, like wind
    - Imperfections in the rocket's geometry, as a result of construction.
- Considerations:
    - Compute costs (number of cells?)
    - Compute time
    - OpenFOAM solver
        - rhoSimpleFoam
        - simpleFoam

4.7.4.1 Coordinate System and Rocket Geometry

4.7.5 Bounding Box

4.7.6 Inlet/Outlet

SimulationVelocity	

4.7.7 Hardware overview

Two custom PCB's will be designed, one for the nosecone pressure ports, and one for the avionics bay. The projected flight computers will feature the following electronic components:

4.7.7.1 Pressure Transducers

Each flight computer will host 6 separate PSA071040 pressure transducers. This transducer was chosen for its high-pressure range, accuracy of +- 0.3, and low cost. It also features an integrated tubeport of 3mm diameter, which the silicone capillary can slide over. This may introduce the need for an O ring or band to keep the seal airtight. At only about $2 per transducer, this option does carry some technical challenges, however. More details are below.

4.7.7.2 Microcontroller

The STM32G4 will serve as the primary microcontroller on each flight computer. It is a 32-bit mixed-signal microcontroller built for advanced analog and power applications. The biggest downside of this microcontroller is the programming complexity associated with STM32 chips.

4.7.7.3 Analog to Digital Converter

The Texas instruments ADS1256 was chosen as the primary ADC. This device was chosen over more modern alternatives due to its ability to host all 6 pressure transducers on one ADC. Other ADC's would be limited to 5 pressure transducers. It samples at up to 30,000 per second, with 24-bit resolution.

4.7.7.4 IMU

The chosen IMU is the Bosch BMI088. This 6-axis transducer was designed for harsh, vibration-heavy environments, making it well suited for applications in rocketry. It supports acceleration ranges up to +-24g, which, for a 10kg rocket should be sufficient. The K455 motor has a peak thrust of 1308N.

$$F = ma$$
$$Thrust + mg = ma$$
$$\frac{1308}{10} - g = a$$

Peak acceleration is calculated to be 12.3g, well within the 24g range, making the sensor safe from clipping.

4.7.8 Software

The central point of the project is a CFD surrogate model which will be given inputs from flight state and a combination of pressure sensors and will predict the pressure at the remaining sensors. The combination will be decided later after full investigation into CFD simulations of the rocket to find the optimal sets or predicted and observed data. In effect:

$$f(p_1, p_2 \dots v, \beta_{alt} \dots) \rightarrow [p_8 \quad p_9 \quad p_{10} \quad p_{11} \quad \dots]$$

To train the first CFD surrogate, a data set will be generated by running CFD using OpenFOAM over the areas of the sub-scale rocket with pressure sensors (the top half, ignoring fins to reduce complexity) under various combinations of altitude, roll, angle of attack, and velocity. Specific ranges for these parameters are to be decided and will be adjusted after analysis of flight data.

The neural network observes some of the pressure data, allowing it to learn patterns in airflow around the rocket, and giving it access information related to wind or angle of attack which otherwise would be difficult to measure and present as clean observations.

After training, the model will be a specialized CFD surrogate for our rocket. By comparing the predictions of the neural network to those observed in real life, we will obtain our first quantifiable results, that being the deviation between CFD and reality.

$$p_{corrected}(p_1, p_2 \dots v, \beta_{alt} \dots) = f(p_1, p_2 \dots v, \beta_{alt} \dots) + [\Delta P_1 \quad \Delta P_1 \quad \dots]$$

The second step would be the training of a correcting neural network on the outputs of the CFD surrogate and the actual results, allowing the investigation of an approach to aerodynamic state estimation that is trained in simulation and only requires small amounts of correcting real-world data.

After the methodology is validated on the sub-scale rocket, it will be applied to the full-scale rocket. A base CFD surrogate will be trained on a new CFD data set with value ranges improved after evaluation of flight data collected during the test flight. A correction model will be trained on the actual read values during the test flight following similar methodology to the sub-scale rocket models.
After the methodology is validated on the sub-scale rocket, it will be applied to the full-scale rocket. A base CFD surrogate will be trained on a new CFD data set with value ranges improved after evaluation of flight data collected during the test flight. A correction model will be trained on the actual read values during the test flight following similar methodology to the sub-scale rocket models.

4.7.9 Technical Challenges and Proposed Solutions

There are many technical challenges that must be addressed in order to properly perform this experiment. Though many more challenges are still to be discovered, some likely struggles can still be addressed:

Surface pressure is impacted by both aerodynamics and altitude:

To properly perform the experiment, aerodynamic pressure changes must be isolated from altitude-related pressure changes. Thus, the avionics bay flight computer would need to average its pressure measurements (to balance out angle of attack's affect) to calculate altitude and therefore calculate the ambient air pressure, which can then be used in conjunction with the individual pressure sensors on the avionics bay and nosecone to understand changes in pressure caused exclusively by factors other than altitude like velocity or angle of attack.

The avionics bay may not have enough space to house all 4 independent circuits:

In the event that the avionics bay is not large enough to house the two altimeters, GPS transmitter, and payload experiment, the switch band and tube coupler may be expanded. This has little negative effects to the rocket and increases general stability.

The PSA071040 outputs a voltage range of 0 to 135mV:

This range means that standard ADCs cannot read its output. To address this issue, an amplifier would need to be used. This also means that any extended wiring between this pressure transducer and the amplifier would have large negative affects on data fidelity. This provides further reason to wire the pressure taps to the micro controller directly as opposed to placing 6 separate pressure transducer boards around the airframe and wiring them electrically to the flight controller.

5 Project Plan

5.1 Development Schedule/Timeline, WBS

The table below shows the development timeline for this year's project.

Table 5.1.1: Season Timeline

Tasks	Start Date	Due Date
Handbook & Request for proposal document is released	August 10, 2026	
Proposal	August 10, 2026	September 14, 2026
Awarded Proposals are announced	October 1, 2026	
Kickoff and PDR Q&A	October 6, 2026	

Preliminary Design Review	October 6, 2026	October 29, 2026
PDR Video Conferences	November 9, 2026	November 24, 2026
Critical Design Review Q&A	December 3, 2026	
Subscale Flight Deadline		January 7, 2027
Critical Design Report	December 3, 2026	January 7, 2027
CDR Video Conferences	January 18, 2027	February 3, 2027
FRR Q&A	February 9, 2027	
Vehicle Demonstration Flight Deadline		March 8, 2027
FRR Report	February 9, 2027	March 8, 2027
FRR Video Conferences	March 15, 2027	March 26, 2027
Payload Demonstration Flight		March 29, 2027
Vehicle Demonstration Re-Flights		March 29, 2027
Launch Week in Huntsville, Alabama	April 7, 2027	April 11, 2027
PLAR Deadline		April 26, 2027

5.2 Budgeting

This year's budget cover a wide range of different areas. Other than the rocket itself, we will need supplies for our payload, workshop, and outreach.

Table 5.2.1: Season Budget Breakdown

Group	Name	Vendor	Quantity	Price /Unit	Total Price
Rocket Structure	5.5" Blue Tube	Apogee	2	82.78	165.56
	54 mm Blue Tube	Apogee	1	23.94	23.94
	Tube Coupler	Apogee	1	74.57	74.57
	Coupler Bulkhead Disk	Apogee	2	9.98	19.96
	Fin Can				
	Motor Mount Tube (5.5 in Blue)	Apogee	1	29.60	29.60
	54mm Motor Retainer	Apogee	2	37.97	75.94
	CR-29 38 mm Plywood Centering Rings	Apogee	1	6.55	6.55
	1515 Rail Button	Apogee	1	7.49	7.49
				Total:	**403.61**
Construction	Bob Smith Industries Slow Cure	Amazon	2	18.98	37.96
	JB Weld SteelWeld Epoxy	Amazon	3	16.95	50.85
	BSI Insta-Cure CA Glue	Amazon	1	9.95	9.95
	BSI Insta-Cure+ CA Glue	Amazon	1	13.99	13.99
	BSI Insta-Set CA Accelerator	Amazon	1	9.99	9.99
				Total:	**122.74**
Recovery	2 Foot Rocketman Drogue Parachute	Rocketman Parachutes	1	5.78	11.56
	72" Rocketman Main Parachute	Rocketman Parachutes	1	65.50	65.50
	4-40 Nylon Sheer Pins (20 Pack)	Apogee	1	7.96	7.96

	Kevlar Cord 1700# (30ft Roll)	Amazon	5	32.95	164.75
	18in Parachute Protector	Apogee	2	16.21	32.42
	Stainless Steel U-Bolt	Home Depot	4	8.24	32.96
	1/4 “ Quick Links	Apogee	2	5.86	11.72
	Nylon Tubular Webbing Tape	Amazon	1	32.95	32.95
	Ejection Canister Caps (2 pack)	Apogee	2	4.51	9.02
				Total:	**368.84**
Avionics	Perfectflite StratoLoggerCF	Perfectflite	2	69.95	139.9
	Screw Switches	Apogee	3	23.28	69.84
				Total:	**209.74**
Payload	M10 400mm Threaded Rods (2 pack)	Amazon	1	16.62	16.62
	5.5” Coupler Bulkhead Disks	Apogee	2	9.98	19.96
	Silicone Tubing 4mm ID 5mm OD	Amazon	1	12.32	12.32
	Copper Tubing 3mm ID 4mm OD	Amazon	1	12.53	12.53
	Pressure Sensor	DigiKey (or other component supplier)	6	2.03	12.18
	PCB Manufacturing	JLCPCB or OSH Park	TBD	TBD	200.00
	Eggfinder GPS Mini Transmitter	Eggtimer Rocketry	2	75	150
				Total:	**423.61**
Outreach	Large Posters	Walgreens	3	25.99	77.97
				Total:	**106.96**
Travel	Round Trip Flights (estimate): SEA->HSV->SEA	TBD	9	900	8100
	Travel to Eastern Washington for High Power Launches	TBD			

				Total:	**8100.00**
Misc.	3D Printer Filament (Potential Discounts available)	Bambu/Amazon	6	22.99	137.94
	Team Merchandise	FRCTees	10	20.00	200.00
				Total:	**337.94**
Total Excluding Travel				**Total:**	**1976.44**
Overall				**Total:**	**10076.44**

5.3 Funding

For funding this year, the team is planning on using $1,000 from the American Rocketry Challenge in which they were rewarded for placing 2nd in the national competition. We also plan on acquiring sponsorships and grants as we reach out to local businesses and companies like Microsoft, Amazon, and Google. We will utilize volunteer-matching from our mentor Mehul who works for Apple. The remaining amount will be distributed and covered evenly between team members.

5.4 STEM Industry OR Community STEM Engagement Plan

5.4.1 Social Media

We will be maintaining three social media platforms through the year: YouTube, Instagram, and a team website. These platforms allow us to extend our reach beyond just our local community and allow us to share our rocketry and engineering knowledge with the broader community.

5.4.2 Community STEM Engagement Plan

Our goal for outreach is to create a lasting and meaningful impact on our community. We want to inspire the next generation by introducing them into the multitude of possibilities that STEM entails. This approach entails both hosting and attending various events, as well as providing a path for students to join rocketry.

Key events:

- Attending Cherry Crest STEM Night to educate parents and elementary schoolers about rocketry opportunities such as WA Youth Rocketry rocket workshops or introduce ARC for the incoming middle schoolers.
- Lunches at various Middle Schools around the Bellevue School District. This is a great way to interact with kids who are ready to join rocketry and participate in ARC.
- Organize a rocketry roundup as a continuation of Odle Middle School's annual rocketry roundup. This is a great opportunity for members from the industry to interact with students to discuss engineering and rocketry.

5.4.3 STEM Industry

We plan to reach out to companies and organizations such as SpaceX, Blue Origin, Honeywell, Boeing and NASA to develop connections that can help support our project through sponsorships, mentoring, and technical help as well. We also intend to connect with CFD experts when working with CFD software to ensure our simulations are accurate and properly configured. These interactions could help us better understand our results.